File Type Identifier

Security Utility v1.0 — Project Documentation

`Magic Number · Extension · Validation`

Project Overview

The File Type Identifier is a browser-based security tool that detects the **true type** of any file by reading its **magic bytes** — the raw binary signature stored in the first 16 bytes — rather than trusting the file extension alone. It compares the claimed extension against the actual detected format and reports whether they match.

This document demonstrates two key scenarios: a legitimate file correctly identified as **SAFE — MATCH**, and a renamed file (extension changed via CMD) flagged as **UNKNOWN**.

Result	Meaning
SAFE — MATCH	Extension matches the detected magic bytes
UNKNOWN	No recognized magic bytes — file may be renamed or plain text
MISMATCH	Extension does not match the actual file format

Scenario 1 — Normal File Identification (SAFE — MATCH)

A genuine JPEG image is uploaded. The tool reads the magic bytes and confirms the extension is legitimate.

Step 1: Upload Interface

The tool opens in the browser with a clean drag-and-drop upload zone. It supports PNG, EXE, JPG, PDF, GIF, ZIP, and more.

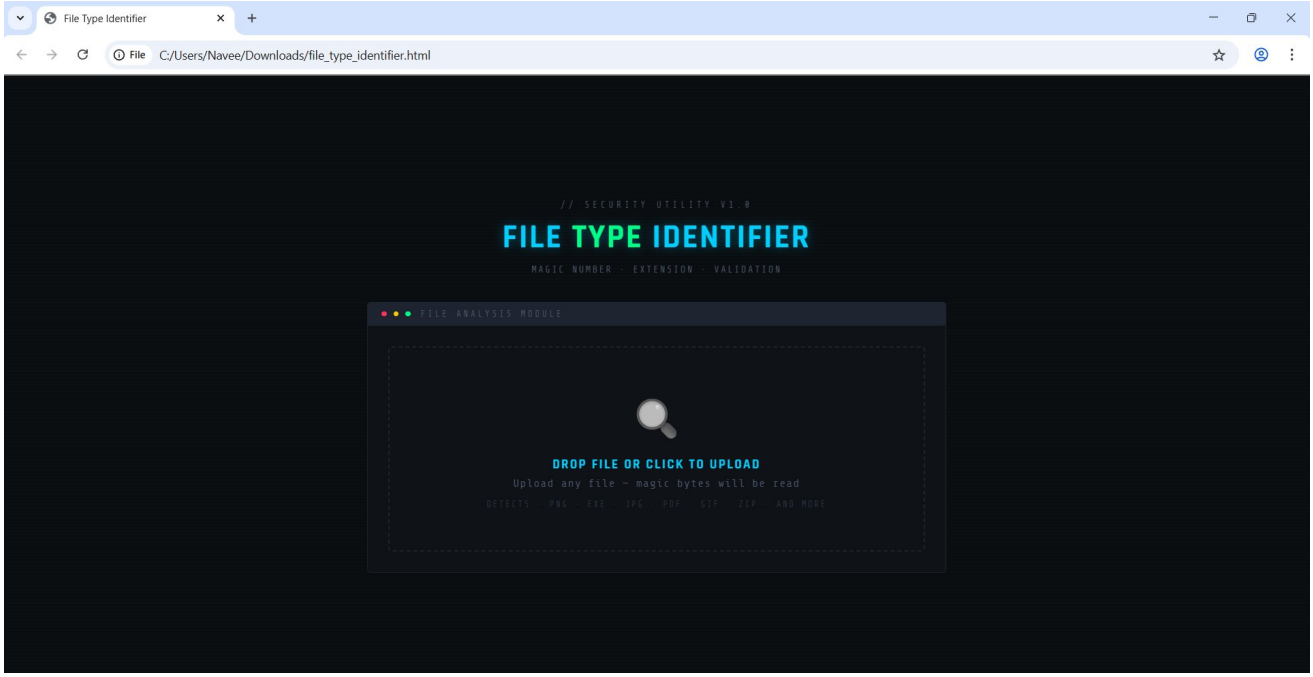

Figure 1 — File Type Identifier landing page with upload zone

Step 2: File Selection

A WhatsApp image (JPEG, ~118 KB) is selected from the Downloads folder. The file extension is .jpeg as saved by WhatsApp.

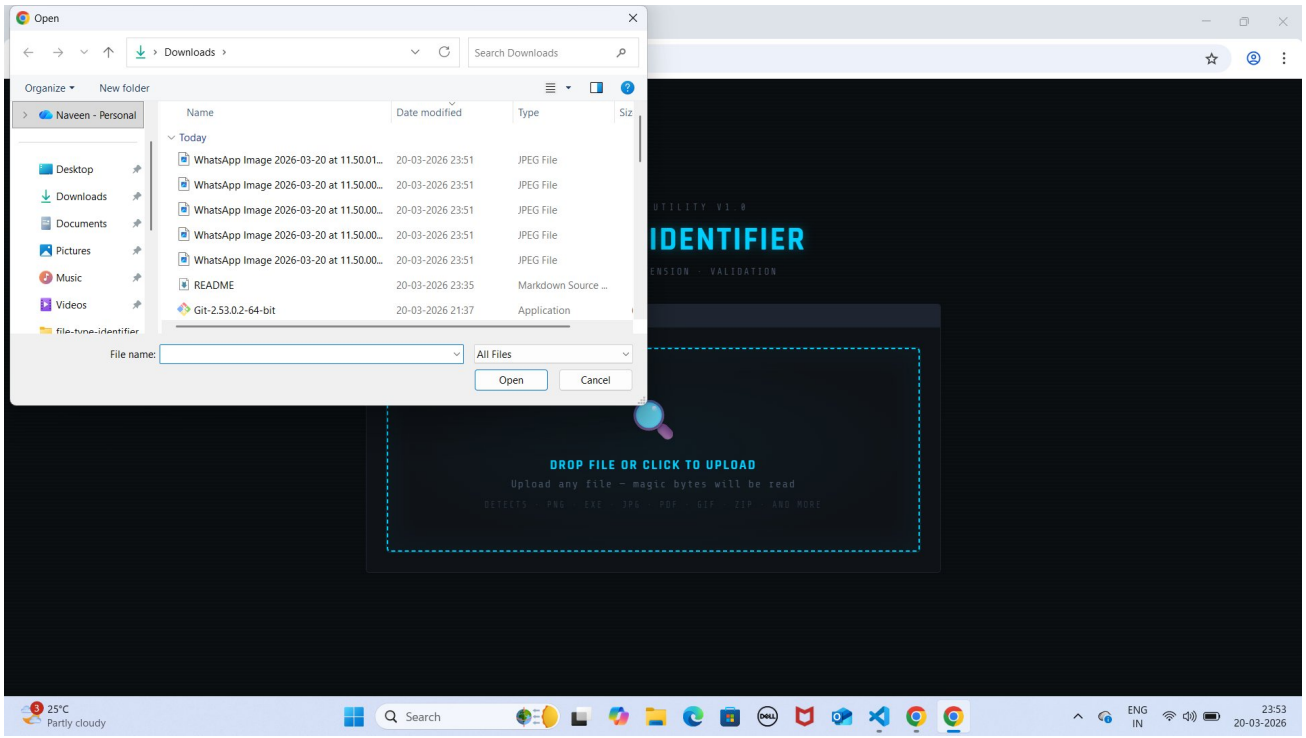

Figure 2 — File browser showing WhatsApp JPEG images in Downloads

Step 3: Result — SAFE — MATCH

After uploading, the tool analyzes the file and displays:

- **Filename:** WhatsApp Image 2026-03-20 at 11.50.01 PM.jpeg
- **File Size:** 118.2 KB
- **Extension (Claimed):** .jpeg
- **Detected Type (Magic):** JPG
- **First 16 Bytes (Hex):** FF D8 FF E0 00 10 4A 46 49 46 00 01 01 00 00 01

The magic bytes **FF D8** are the standard JPEG signature. The extension .jpeg matches the detected type JPG — the tool shows **SAFE — MATCH**, confirming the file is exactly what it claims to be.

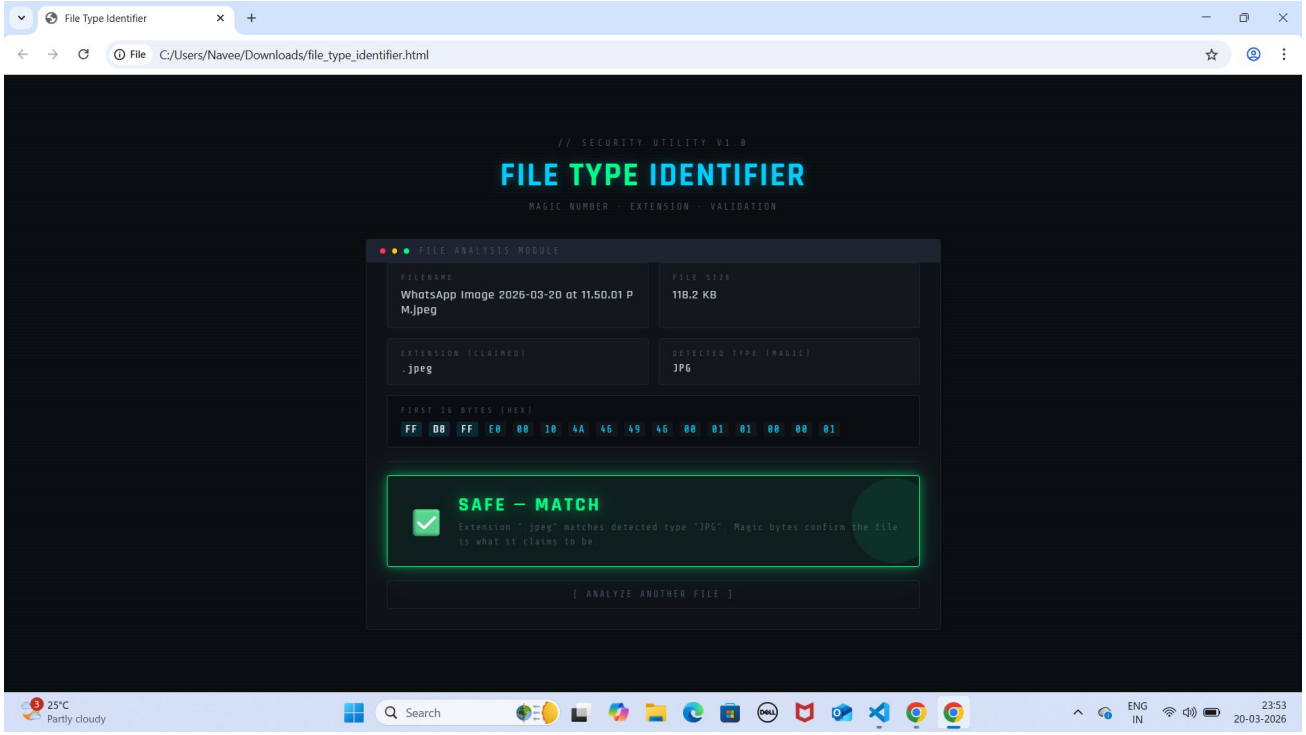

Figure 3 — SAFE — MATCH result: extension .jpeg matches detected type JPG

Scenario 2 — Renamed File Detection (UNKNOWN)

A file is renamed using the **ren** command in CMD. The extension changes, but the internal binary content stays the same. The tool catches it.

Step 1: Rename File Extension via CMD

In Command Prompt, the user navigates to the Documents folder and renames the file from A.png to A.jpg. The rename succeeds — but the file's internal binary content remains unchanged.

```
C:\Users\Navee>cd Documents
C:\Users\Navee\Documents>ren "A.png" "A.jpg"
```

```
C:\Users\Navee>cd Documents

C:\Users\Navee\Documents>ren "A.txt" "A.png"
The system cannot find the file specified.

C:\Users\Navee\Documents>ren "A.png" "A.jpg"

C:\Users\Navee\Documents>
```

Figure 4 — CMD showing ren command renaming A.png to A.jpg

Step 2: File Properties & Upload

The file properties show A.jpg as a "JPG File" with a size of **0 bytes** — no real image data inside. The file browser confirms it is the only file in the Documents folder.

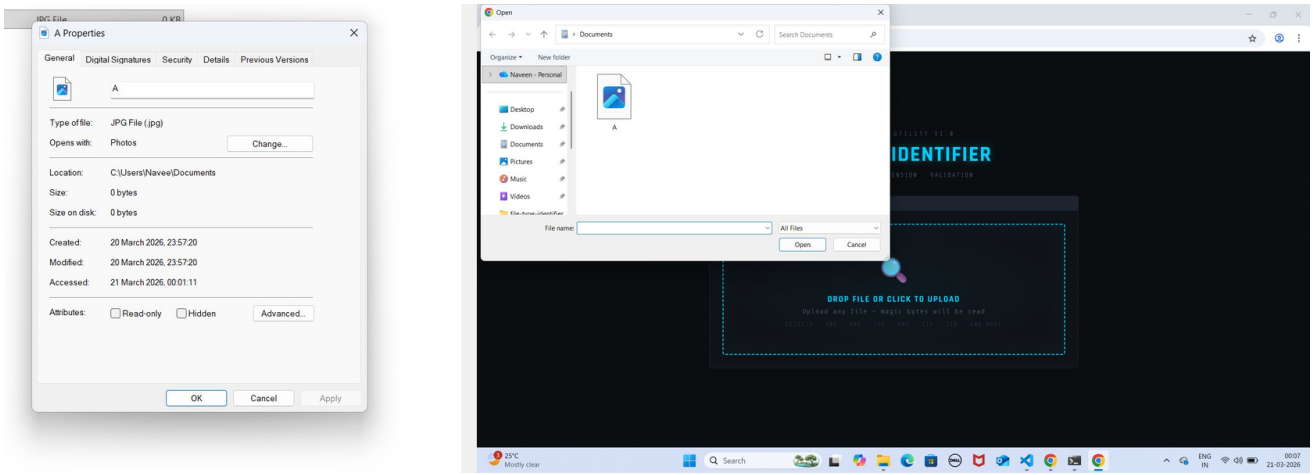

Figure 5 — Left: A.jpg properties showing 0 bytes | Right: file browser selecting A.jpg for upload

Step 3: Result — UNKNOWN

The tool reads the first 16 bytes. No recognized magic bytes are found. The detected type is reported as **Unknown / Plain Text** and the tool raises an **UNKNOWN** warning — proving the file was renamed and is not a genuine JPEG.

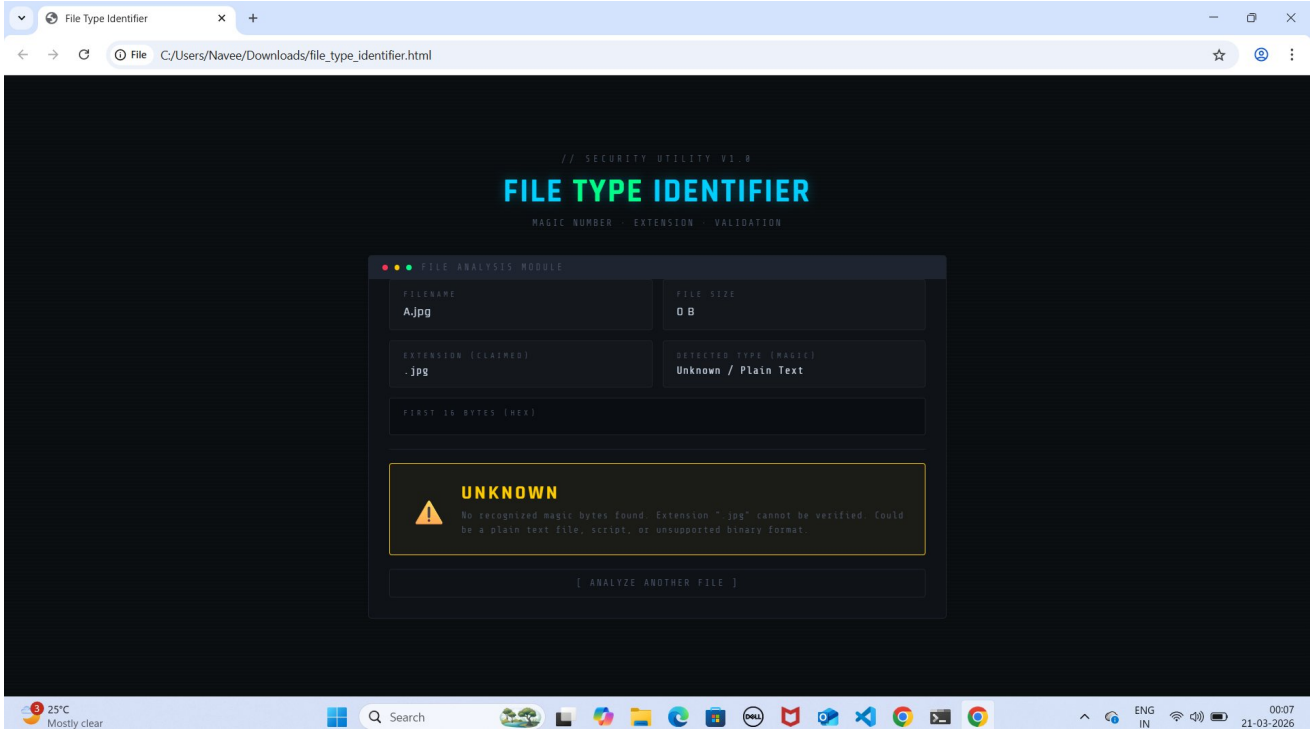

Figure 6 — UNKNOWN result: no magic bytes found, extension .jpg cannot be verified

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Key Takeaway

Renaming a file's extension — whether via CMD, File Explorer, or any other method — **never** changes the actual binary content of the file. The File Type Identifier reads the raw magic bytes to determine the real format, making it impossible to disguise a file by simply changing its extension.

> *"A file is what its bytes say it is — not what its name says."*